Kubernetes en Production

Kubernetes Production · Lesson 0

One path: **build** clusters with **Kubespray**, **operate** them with admin skills (**CKA**)

Welcome

This formation **merges** the CKA administration course with the Kubespray deployment course into **one coherent path** — from install to Day-2.

The big idea:

- **Build** the cluster with **Kubespray** (production-grade, Ansible-driven)
- **Operate & administer** it with core admin skills (**CKA scope**)
- `kubeadm` is kept only as the *under-the-hood* mechanism Kubespray drives

> Goal: not just pass an exam, but **build and run** production Kubernetes with confidence.

Learning Objectives

By the end of this lesson you will be able to:

- Describe **what Kubernetes is** — the production container platform
- Name the **control plane** and **worker** components at a glance
- Explain this course's promise: **build with Kubespray + operate (CKA scope)**
- Justify **why Kubespray** for installation vs manual `kubeadm`, kops, managed K8s
- Recognise that **Kubespray uses kubeadm under the hood**
- Map the **four phases** and the **24-lesson roadmap**

> Reference: *Kubernetes overview* — kubernetes.io/docs · *Kubespray* — kubespray.io

0.1 What Is Kubernetes?

Kubernetes — The Production Container Platform

Kubernetes (K8s) is an open-source platform for **automating the deployment, scaling and management** of containerized applications.

It takes a fleet of machines and makes it behave like **one resilient computer**:

- **Schedules** containers (grouped into **Pods**) across the nodes
- **Self-heals** — restarts failed containers, reschedules off dead nodes
- **Scales** workloads up and down on demand
- **Rolls out** changes and **rolls back** on failure
- **Exposes & load-balances** services, manages config and secrets

> It is the de-facto standard for running containers in production.

Control Plane + Workers at a Glance

A cluster has a **control plane** (the brain) and **worker nodes** (the muscle).

Plane	Component	Role
Control plane	`kube-apiserver`	The front door — the cluster API
	`etcd`	Key-value store of all cluster state
	`kube-scheduler`	Places Pods onto nodes
	`kube-controller-manager`	Reconciles desired vs actual state
Worker node	`kubelet`	Node agent — runs the Pods
	container runtime	containerd / CRI-O — runs containers
	`kube-proxy`	Service networking on the node

> We dissect each component in **Lesson 1 — Architecture**.

Declarative & Self-Healing

You don't issue step-by-step commands — you **declare the desired state**.

```yaml
# "I want 3 replicas of nginx, always"
apiVersion: apps/v1
kind: Deployment
spec:
  replicas: 3
  template:
    spec:
      containers:
        - name: web
          image: nginx:1.27
```

1. You `apply` YAML → the **API server** stores it in `etcd`.
2. **Controllers** continuously **reconcile** reality to match the spec.
3. A Pod dies? The controller makes a new one. No human in the loop.

> This *reconciliation loop* is the heart of how Kubernetes operates.

0.2 The Course Promise

One Path: Build It, Then Run It

This is **one** formation, not two stitched together. The redundancy between a CKA course and a Kubespray course has been removed.

You will…	…with
Build production clusters	**Kubespray** (Ansible playbooks)
Operate & administer them	core admin skills (**CKA scope**)
Go from **install → Day-2**	one continuous numbering (00–23)

The biggest change vs a classic CKA path: cluster **creation is taught with Kubespray** instead of a hand-cranked `kubeadm` walkthrough.

> kubeadm isn't dropped — it's the engine Kubespray drives (more on the next section).

Build vs Run — Two Skill Sets, One Cluster

```
Build the cluster  →  Use the cluster  →  Operate it (Day-2)  →  Advanced
(Kubespray)           (CKA app/admin)      (Kubespray + CKA)       (Kubespray)
```

- **Build (Kubespray / Ansible):** stand up and maintain the **fleet** — nodes, runtime, CNI, HA, upgrades.
- **Run (CKA / `kubectl`):** deploy and administer the **workloads** — apps, storage, scheduling, RBAC, troubleshooting.

> Same cluster, two complementary lenses. You leave able to do **both**.

0.3 Why Kubespray for Installation

Why Kubespray (vs the Alternatives)

Approach	Model	Trade-off
Manual `kubeadm`	Bootstraps one cluster; you hand-wire OS prep, CNI, HA	Educational, **not repeatable** at scale
kops	Provisions the infra itself (mostly AWS/GCP)	**Single-cloud**, couples infra + cluster
Managed (EKS/AKS/GKE)	Cloud runs the control plane	You **never learn to operate** it
Kubespray	Ansible drives kubeadm across the **whole fleet** + OS prep + CNI + add-ons + Day-2	**Production, multi-node, multi-platform, repeatable**

> Kubespray = a CNCF / `kubernetes-sigs` project that deploys a **production-ready** cluster with Ansible.

Kubespray's Edge

- **Repeatable & idempotent** — the cluster is described in an **inventory** + **group_vars**; re-runs change only what's needed.
- **HA out of the box** — ≥ 3 control-plane / etcd nodes, load balancing, kube-vip.
- **Multi-OS** — Ubuntu, Debian, RHEL/Rocky/Alma, Flatcar, openSUSE, …
- **Choice of runtime & CNI** — containerd (default) / CRI-O · Calico (default) / Cilium / Flannel / Kube-OVN.
- **Day-2 playbooks** — `scale.yml`, `upgrade-cluster.yml`, backup & recovery, `reset.yml`, air-gapped.

> One Ansible codebase covers the **full lifecycle**: install + scale + upgrade + recover.

Bridge: Kubespray Uses kubeadm Under the Hood

Choosing Kubespray does **not** mean throwing away kubeadm knowledge.

- Since **v2.3**, Kubespray calls **`kubeadm`** **internally** to create the control plane and join nodes.
- You get **kubeadm's correctness** with **Ansible's fleet management**.
- So every kubeadm concept still maps — `init` / `join`, the control-plane components, the control-plane taint, version-skew rules — and shows up again in later lessons.

> Kubespray automates *what you'd otherwise script around kubeadm by hand.*

0.4 How the Course Is Structured

The Four Phases

```
Build the cluster  →  Use the cluster  →  Operate it (Day-2)  →  Advanced
(Kubespray)           (CKA app/admin)      (Kubespray + CKA)      (Kubespray)
```

Phase	What you do	Lessons
Build	requirements → inventory → group_vars → runtime/CNI → `cluster.yml` → HA	02–07
Use	workloads, storage, access, config, scheduling, networking, security, CRDs	08–15
Operate	maintenance & scaling, upgrades, backup/recovery, add-ons, troubleshooting	16–20
Advanced	hardening & offline, cloud & OpenStack, exam prep & wrap-up	21–23

Course Roadmap (1/2) — Build & Use

#	Lesson	Phase
01	Kubernetes Architecture	Build
02	Requirements & Environment	Build
03	Installation & Inventory (Kubespray)	Build
04	Cluster Configuration (group_vars)	Build
05	Container Runtimes & CNI	Build
06	Deploying the Cluster (Day-1)	Build
07	High Availability	Build
08–11	Apps · Storage · Access · Config & Quotas	Use

Course Roadmap (2/2) — Use, Operate & Advanced

#	Lesson	Phase
12–15	Scheduling · Networking · Security · CRDs & Operators	Use
16	Node Maintenance & Scaling	Operate
17	Upgrades	Operate
18	Backup, Recovery & Reset	Operate
19	Add-ons & Packaging (Helm / Kustomize)	Operate
20	Logging, Monitoring & Troubleshooting	Operate
21	Hardening & Offline	Advanced
22	Cloud & OpenStack	Advanced
23	Exam Preparation & Wrap-up	Advanced

0.5 Who This Is For

Who This Is For

Audience

- Sysadmins, DevOps and platform engineers running or planning to run Kubernetes
- Anyone targeting the **CKA** certification

Prerequisites

- Comfortable on the Linux command line
- Basic containers (containerd/Docker, images, registries) and **YAML**
- Networking fundamentals (IP, DNS, ports); a little **Ansible** helps for the build

> **No prior Kubernetes experience required** — we start from the architecture.

What You'll Be Able to Do

By the end of the formation you can:

- **Build** a production-grade, **HA** cluster with Kubespray — bare metal or cloud (incl. **OpenStack**)
- **Deploy & expose** applications; manage **storage, scheduling, config, RBAC**
- Run **Day-2 operations** — scale, upgrade, back up, recover, troubleshoot
- Operate with the **coverage of the CKA exam** and the **repeatability** of Ansible automation

> Build it, run it, keep it running — that's the whole journey.

Key Takeaways

- **Kubernetes** = the production container platform: **control plane** (brain) + **workers** (muscle), driven by **declarative reconciliation**.
- This course is **one path**: **build with Kubespray**, **operate with CKA-scope admin skills** — install to Day-2.
- **Kubespray** wins for production install: **repeatable, HA, multi-OS, CNI/runtime choice, Day-2 playbooks** — and it **uses kubeadm under the hood**.
- The journey runs in **four phases** — **Build** → **Use** → **Operate** → **Advanced** — across **24 lessons**.
- No K8s experience needed — we start at the **architecture**.

End of Lesson 0

Next: Lesson 1 — Kubernetes Architecture

Build with Kubespray · run with `kubectl` · `kubeadm` underneath